

AI MENTAL HEALTH

ANA – 1 (WEEK-4)

MHSDS KPI Extraction Report

**Dataset Used: Mental Health Services Monthly Statistics,
Performance March 2026**

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1. Introduction

The Mental Health Services Data Set (MHSDS) is a national NHS dataset used to monitor mental health service activity, patient pathways, hospital admissions, crisis response services, and referral-to-treatment performance across England.

This project focuses on extracting five major NHS Mental Health Key Performance Indicators (KPIs) from the MHSDS datasets and building a source mapping table for performance analysis and reporting.

The extracted KPIs include:

- Beds
- Contacts
- Crisis Calls
- Admissions
- Referral to Treatment (RTT)

2. Objectives

The primary objectives of this project were:

- Extract NHS mental health KPIs from MHSDS datasets
- Analyze RTT and waiting-time indicators
- Identify NHS KPI metric IDs
- Build KPI summary tables
- Create source mapping documentation
- Produce KPI visualizations
- Generate operational reporting outputs

3. Datasets Used

Dataset	Purpose
MHSDS Data_MarPerf_2026_v2.csv	Main KPI extraction dataset
MHSDS Data_Rstr_Mar2026_Prfl.csv	Restricted NHS metrics
MHSDS Time_Series_data_Apr2025_Mar_Perf_2026_v2.csv	Historical KPI validation
RTT waiting times.xlsx	RTT and waiting-time analysis
Data Coverage File CSV	Data completeness assessment
Data Quality VODIM and Integrity File	Data quality validation

4. Technologies Used

Technology	Purpose
Python	Data analysis
Pandas	Data manipulation
NumPy	Numerical processing
Matplotlib	Data visualization
Google Colab	Workflow execution
CSV / Excel	NHS data storage

5. Methodology

5.1 Data Loading

The NHS datasets were imported using Python libraries including:

- pandas
- numpy
- matplotlib

CSV and Excel datasets were loaded separately according to their file structures.

```
# Main performance dataset
mhsds_perf = pd.read_csv(
    "/content/MHSDS_Data_MarPerf_2026_v2.csv"
)

# Restricted dataset
mhsds_rstr = pd.read_csv(
    "/content/MHSDS_Data_Rstr_Mar2026_Prfl.csv"
)

# Time series dataset
mhsds_time = pd.read_csv(
    "/content/MHSDS_Time_Series_data_Apr2025_Mar_Perf_2026_v2.csv"
)

# RTT Excel dataset
rtt_data = pd.read_excel(
    "/content/RTT_waiting_times.xlsx"
)
```

5.2 Data Cleaning

Several preprocessing steps were performed:

- Column name standardization
- Invalid value removal
- Suppressed NHS value handling
- Numeric conversion
- Missing value handling
- Header row detection

Suppressed NHS values represented by * were converted to NaN before analysis.

6. KPI Identification

Relevant KPI metrics were identified using:

- MEASURE_NAME
- MEASURE_ID
- RTT metric identifiers

Keywords used included:

- bed
- contact
- crisis
- admission
- RTT

7. KPI Extraction

The KPI extraction process used aggregation techniques including:

- Sum
- Mean
- Median
- Maximum
- Minimum

KPIs were extracted using predefined NHS metric IDs.

8. KPI Categories

8.1 Beds KPI

Description

The Beds KPI measures mental health inpatient bed activity and occupied bed-day utilization.

Measure IDs

Measure ID	Description
MHS24	Bed days in reporting period
MHS25	Bed days less leave
MHS156	Adult acute and PICU discharges

Analysis

The Beds KPI identified inpatient mental health service utilization across NHS providers and organizations.

8.2 Contacts KPI

Description

Measures community mental health care contacts and service activity.

Measure IDs

Measure ID	Description
MHS29d	Community mental health contacts

MHS30f	Attended contacts
MH01	People in contact with services

Analysis

This KPI reflects patient engagement and mental health service activity levels.

8.3 Crisis Calls KPI

Description

Measures urgent mental health crisis activity and emergency interactions.

Measure IDs

Measure ID	Description
CCR117	Very urgent crisis referrals
CCR118	Crisis face-to-face contacts
CCR119	Crisis referrals within 4 hours
CCR120	Urgent referrals within 24 hours

Analysis

Crisis KPIs provide operational insight into urgent mental health demand.

8.4 Admissions KPI

Description

Measures inpatient psychiatric admissions activity.

Measure IDs

Measure ID	Description
MHS27	Hospital admissions
MHS106	Acute admissions
MHS106a	Acute admissions with previous MH contact
MHS106b	Acute admissions without previous MH contact

Analysis

Admissions KPIs reflect acute mental health care utilization and inpatient demand.

8.5 RTT KPI

Description

Measures NHS referral-to-treatment waiting-time performance.

RTT KPI Metrics

KPI	Metric ID
People Entering RTT Pathway	EIP23a
People Waiting	EIP23c
People Treated	EIP23b
RTT Performance Percentage	EIP23j
Average Waiting Time	EIP67

9. Exploratory Data Analysis (EDA)

9.1 Data Structure

The RTT dataset contained:

- Multiple sheets
- Provider-level metrics
- ICB-level metrics
- RTT percentages

- Waiting-time indicators

9.2 Data Quality Considerations

Several NHS data quality issues were identified:

- Missing values
- Suppressed low-count values
- Inconsistent RTT structures
- Organization-level aggregation

These issues were resolved during preprocessing.

10. KPI Source Mapping Table

KPI	Description
Beds	Occupied mental health bed activity
Contacts	Mental health care contacts
Crisis Calls	Crisis mental health activity
Admissions	Mental health admissions
RTT	Referral-to-treatment metrics

A1				KPI_CATEGORY	
	A	B	C	D	E
1	KPI_CATEGORY	VALUE			
2	Admission	325138			
3	Beds	4254062			
4	Contacts	26378739			
5	Crisis Calls	67990			
6	RTT	176311			
7					
8					
9					
10					

KPI_Categ	Measure_ID	Description	Source_Dataset		
Beds	MHS24, MI	Mental health	MHSDS Data_MarPerf_2026_v2.csv		
Contacts	MHS29d, MI	Community	MHSDS Data_MarPerf_2026_v2.csv		
Crisis Calls	CCR117, C	Mental health	MHSDS Data_MarPerf_2026_v2.csv		
Admission	MHS27, MI	Psychiatric	MHSDS Data_MarPerf_2026_v2.csv		
RTT	MHS125, MI	Referral to RTT waiting times.xlsx			

11. Source Mapping Documentation

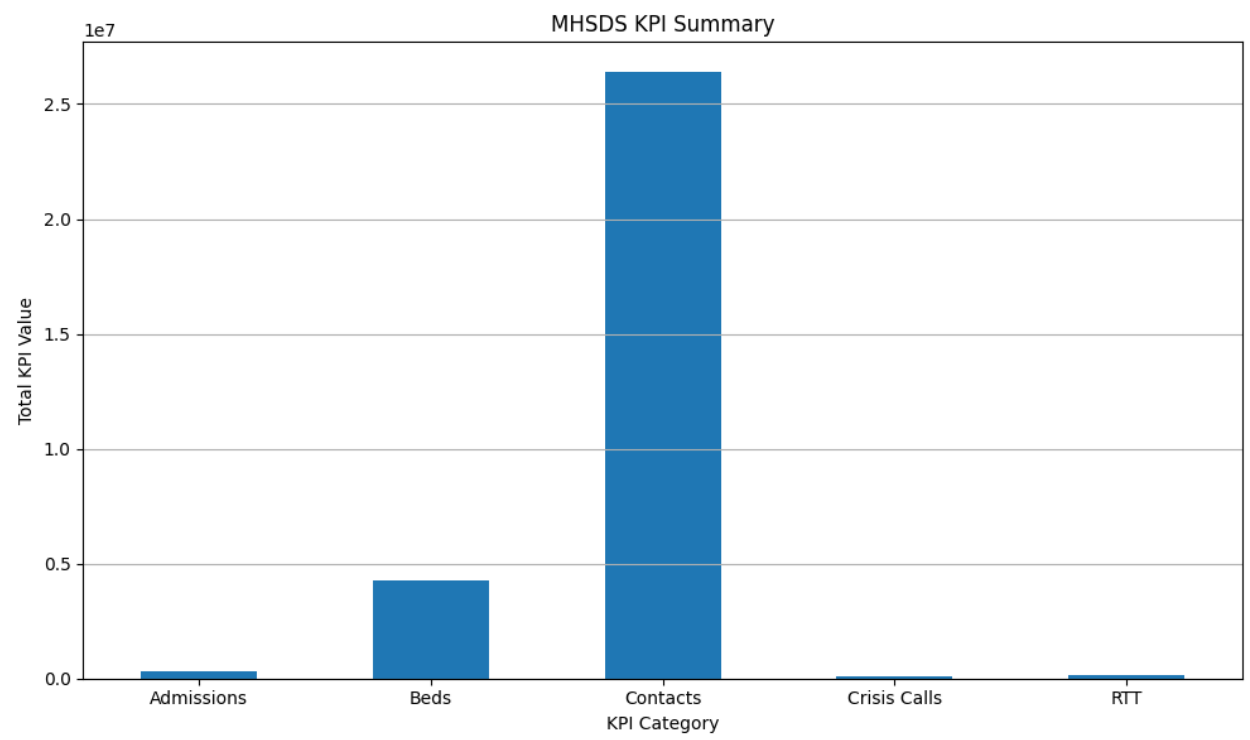
KPI	Metric ID	Source File	Source Sheet
Beds	Keyword: bed	MHSDS Data_MarPerf_2026_v2	Main dataset

Contacts	Keyword: contact	MHSDS Data_MarPerf_2026_v2	Main dataset
Crisis Calls	Keyword: crisis	MHSDS Data_MarPerf_2026_v2	Main dataset
Admissions	Keyword: admission	MHSDS Data_MarPerf_2026_v2	Main dataset
RTT Performance	EIP23j	RTT waiting times	Table 2a
People Waiting	EIP23c	RTT waiting times	Table 2a
People Treated	EIP23b	RTT waiting times	Table 2a
Average Waiting Time	EIP67	RTT waiting times	Table 5a

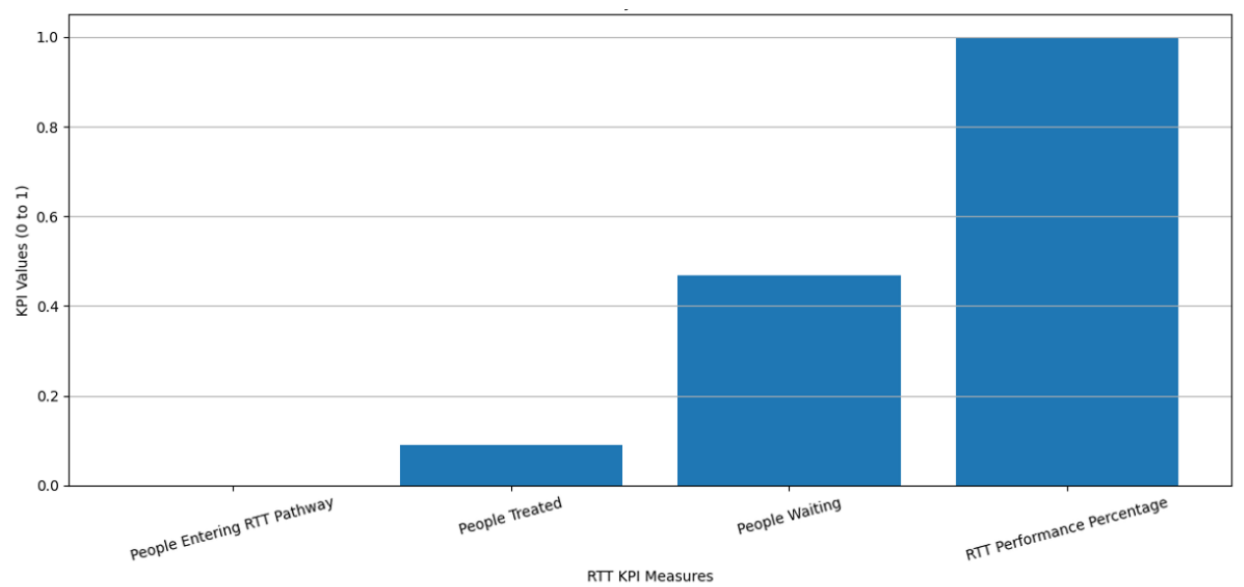
12. Visualizations

The following visualizations were created:

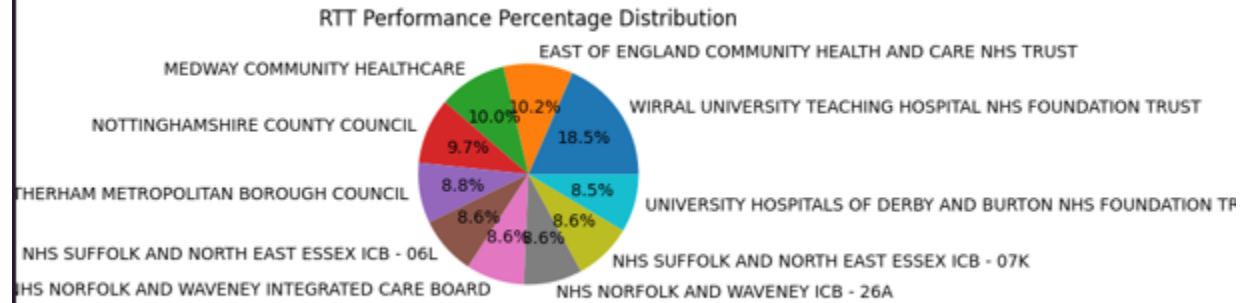
1. MHSDS KPI Bar Chart



2. RTT KPI Summary Chart



3. RTT Performance Percentage Pie Chart



13. Key Findings

13.1 Structured NHS KPI Framework

The MHSDS dataset uses a structured KPI system through:

- MEASURE_ID
- MEASURE_NAME
- MEASURE_VALUE

13.2 RTT Performance Complexity

The RTT dataset contains:

- Multiple reporting layers
- Provider and ICB metrics
- Access standards
- Waiting-time percentages

13.3 Operational Reporting Value

The extracted KPIs support:

- NHS operational reporting
- Service-demand monitoring
- RTT assessment
- Mental health service planning

14. Conclusion

This project successfully extracted and analyzed NHS Mental Health KPIs from MHSDS and RTT datasets.

The project demonstrated how NHS performance data can be transformed into:

- KPI extraction tables
- RTT performance summaries
- Source mapping documentation
- Operational visualizations
- NHS reporting outputs

The framework provides a scalable approach for future NHS operational analytics and mental health performance reporting.